

作业题 15

题目

[作业 15]: 设体系的能量为 $H(x, y) = -(x^2 + y^2) + \frac{1}{2}(x^4 + y^4) + \frac{1}{3}(x - y)^4$,
取 $\beta = 0.2, 1, 5$, 采用 Metropolis 抽样法计算 $\langle x^2 \rangle, \langle y^2 \rangle, \langle x^2 + y^2 \rangle$ 。
抽样时在 2 维平面上依次标出 Markov 链点分布, 从而形象地理解 Markov 链。

理论计算:

细致平衡条件: $\frac{S(i)}{S(j)} = \frac{G(i|j) * A(i|j)}{G(j|i) * A(j|i)}$
选取 $A(j|i) = \min(1, \frac{S(j) * G(i|j)}{S(i) * G(j|i)})$ 满足上述条件
此处 $G(i|j)$ 可以任选一个合适分布,
此题选择以 j 为中心, 长度为 $2d$ 的均匀分布,
故 $G(i|j) = G(j|i)$
而 $S(i) = ce^{-\beta H_i}, \beta$ 代表温度
故 $\frac{S(i)}{S(j)} = e^{\beta(H_j - H_i)}$

算法

Metropolis-Hastings(x_0, P, Q)

For $k = 0, \dots, N - 1$ do:

1. Sample x' from $Q(x'|x_k)$.
2. With acceptance probability $\alpha(x', x_k) = \min\left(1, \frac{P(x')Q(x_k|x')}{P(x_k)Q(x'|x_k)}\right)$, set $x_{k+1} \leftarrow x'$.
3. Otherwise, set $x_{k+1} \leftarrow x_k$.

代码

```
1. import numpy as np
2. # import math
3. import matplotlib.pyplot as plt
4. # from concurrent.futures import ThreadPoolExecutor
5.
6.
7.
8. aaaaaaaaa = 16807
9. m = 2147483647
10.
11. def azmodm(z):
12.     global m, aaaaaaaaa
13.     q = int(m / aaaaaaaaa)
14.     r = m % aaaaaaaaa
15.     tmp = aaaaaaaaa * (z % q) - r * int(z / q)
16.     if tmp >= 0:
17.         return tmp
18.     else:
19.         return tmp + m
20.
21. bbbssbbb = 0
22. aaasssaaa = 0
23. cccssccc = 0
24.
25. def random01():
26.     global bbbssbbb, aaasssaaa, m, cccssccc
27.     z1 = 666955
28.     z2 = 816815
29.     z3 = 116560
30.     z4 = 4813144
31.     if aaasssaaa == 0 and cccssccc == 0:
32.         bbbssbbb = z1
33.     if aaasssaaa == m - 10000000 and cccssccc == 0:
34.         bbbssbbb = z2
35.         aaasssaaa = 1
36.         cccssccc = 1
37.     if aaasssaaa == m - 10000000 and cccssccc == 1:
38.         bbbssbbb = z3
39.         aaasssaaa = 1
40.         cccssccc = 2
41.     if aaasssaaa == m - 10000000 and cccssccc == 2:
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42.         bbbssssbbb = z4
43.         aaasssaaa = 1
44.         cccssssccc = 3
45.         bbbssssbbb = azmodm(bbbssssbbb)
46.         aaasssaaa += 1
47.         return bbbssssbbb / m
48.
49.
50. def H(xy):
51.     x=xy[0]
52.     y=xy[1]
53.     return -(x**2+y**2)+0.5*(x**4+y**4)+1/3*(x-y)**4
54.
55. def P(H0,H1,bt):
56.     return min(1,np.exp(-bt*(H1-H0)))
57.
58.
59. def addone():
60.     global xy1,xy2,xy3
61.     a=random01()
62.     b=random01()
63.     # dx=np.sqrt(-2*np.log(a))*np.cos(2*np.pi*b)
64.     # dy=np.sqrt(-2*np.log(a))*np.sin(2*np.pi*b)
65.     dx=(a-0.5)
66.     dy=(b-0.5)
67.
68.     bt=0.2
69.     x=xy1[-1][0]+dx*d
70.     y=xy1[-1][1]+dy*d
71.     H0=H(xy1[-1])
72.     H1=H([x,y])
73.     c=random01()
74.     p=P(H0,H1,bt)
75.     if p>=c:
76.         xy1.append([x,y])
77.     else :
78.         xy1.append(xy1[-1])
79.         # print(1)
80.
81.     bt=1
82.     x=xy2[-1][0]+dx*d
83.     y=xy2[-1][1]+dy*d
84.     H0=H(xy2[-1])
85.     H1=H([x,y])

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86.     c=random01()
87.     p=P(H0,H1,bt)
88.     if p>=c:
89.         xy2.append([x,y])
90.     else :
91.         xy2.append(xy2[-1])
92.         # print(1)
93.
94.     bt=5
95.     x=xy3[-1][0]+dx*d
96.     y=xy3[-1][1]+dy*d
97.     H0=H(xy3[-1])
98.     H1=H([x,y])
99.     c=random01()
100.    p=P(H0,H1,bt)
101.    if p>=c:
102.        xy3.append([x,y])
103.    else :
104.        xy3.append(xy3[-1])
105.        # print(1)
106.
107.
108.    i=0
109.    j=0
110.    # x=0
111.    # c=1    ##初始的 c 值
112.    N=10000
113.    d=2
114.
115.
116.
117.    xy1=[]
118.    xy1.append([random01(),random01()])
119.    xy2=[]
120.    xy2.append([random01(),random01()])
121.    xy3=[]
122.    xy3.append([random01(),random01()])
123.    # # lam2=1/E
124.
125.    x61=[]
126.    y61=[]
127.    x62=[]
128.    y62=[]
129.    x63=[]

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130.     y63=[]
131.     x6jiay61=[]
132.     x6jiay62=[]
133.     x6jiay63=[]
134.
135.
136.     plt.ion() # 开启交互模式
137.     figure,axes=plt.subplots(2,3,figsize=(12,7))
138.     ax = axes.flatten() #子图展平,将 ax 由 n*m 的 Axes 组展平成 1*nm 的 Axes 组(二维变一
    维)
139.
140.
141.     for k in range(N):
142.         addone()
143.         # n=len(xy)
144.         # print(n)
145.         # plt.clf() # 清空画布
146.         xy1=np.array(xy1)
147.         xy2=np.array(xy2)
148.         xy3=np.array(xy3)
149.         ax[0].cla() # 清空画布
150.         ax[1].cla()
151.         ax[2].cla()
152.         ax[3].cla() # 清空画布
153.         ax[4].cla()
154.         ax[5].cla()
155.
156.         ax[0].set_xlabel(" $\beta=0.2$ ")
157.         ax[1].set_xlabel(" $\beta=1$ ")
158.         ax[2].set_xlabel(" $\beta=5$ ")
159.
160.         # ax[0].set_xlim(min(xy1[:,0])-2,max(xy1[:,0])+2)
161.         # ax[0].set_ylim(min(xy1[:,1])-2,max(xy1[:,1])+2)
162.         x, y = xy1[:,0],xy1[:,1]
163.         ax[0].scatter(x, y,s=1)
164.
165.         # ax[1].set_xlim(min(xy2[:,0])-2,max(xy2[:,0])+2)
166.         # ax[1].set_ylim(min(xy2[:,1])-2,max(xy2[:,1])+2)
167.         x, y = xy2[:,0],xy2[:,1]
168.         ax[1].scatter(x, y,s=1)
169.
170.         # ax[2].set_xlim(min(xy3[:,0])-2,max(xy3[:,0])+2)
171.         # ax[2].set_ylim(min(xy3[:,1])-2,max(xy3[:,1])+2)
172.         x, y = xy3[:,0],xy3[:,1]

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173.         ax[2].scatter(x, y,s=1)
174.
175.
176.         n=k+2
177.         x61.append([n,sum((xy1[:,0])**2)/n])
178.         y61.append([n,sum((xy1[:,1])**2)/n])
179.         x6jiay61.append([n,x61[k][1]+y61[k][1]])
180.         x62.append([n,sum((xy2[:,0])**2)/n])
181.         y62.append([n,sum((xy2[:,1])**2)/n])
182.         x6jiay62.append([n,x62[k][1]+y62[k][1]])
183.         x63.append([n,sum((xy3[:,0])**2)/n])
184.         y63.append([n,sum((xy3[:,1])**2)/n])
185.         x6jiay63.append([n,x63[k][1]+y63[k][1]])
186.
187.         x61 = np.array(x61)
188.         y61 = np.array(y61)
189.         x6jiay61 = np.array(x6jiay61)
190.         x62 = np.array(x62)
191.         y62 = np.array(y62)
192.         x6jiay62 = np.array(x6jiay62)
193.         x63 = np.array(x63)
194.         y63 = np.array(y63)
195.         x6jiay63 = np.array(x6jiay63)
196.
197.         for i in [3,4,5]:
198.             ax[i].set_xlabel("number")
199.             ax[i].set_ylabel("value")
200.             x, y = [x61[:, 0], x62[:, 0], x63[:, 0]][i-
201.                 3], [x61[:, 1], x62[:, 1], x63[:, 1]][i-3]
202.             ax[i].plot(x,y,label=" $x^2$ ")
203.             x, y = [y61[:, 0], y62[:, 0], y63[:, 0]][i-
204.                 3], [y61[:, 1], y62[:, 1], y63[:, 1]][i-3]
205.             ax[i].plot(x,y,label=" $y^2$ ")
206.             x, y = [x6jiay61[:, 0],x6jiay62[:, 0], x6jiay63[:, 0]][i-
207.                 3], [x6jiay61[:, 1], x6jiay62[:, 1], x6jiay63[:, 1]][i-3]
208.             ax[i].plot(x,y,label=" $x^2+y^2$ ")
209.             ax[i].legend(  fontsize='small')
210.
211.
212.         x61 = x61.tolist()
213.         y61 = y61.tolist()
214.         x6jiay61 = x6jiay61.tolist()
215.         x62 = x62.tolist()

```

```

214.     y62 = y62.tolist()
215.     x6jiay62 = x6jiay62.tolist()
216.     x63 = x63.tolist()
217.     y63 = y63.tolist()
218.     x6jiay63 = x6jiay63.tolist()
219.     xy1=list(xy1)
220.     xy2=list(xy2)
221.     xy3=list(xy3)
222.     plt.pause(0.000000000001)
223.
224.     plt.ioff()
225.     print("d={0}".format(d))
226.     print("β=0.2,<x^2>={0},<y^2>={1},<x^2+y^2>={2}".format(x61[N-1][1],y61[N-
1][1],x6jiay61[N-1][1]))
227.     print("β=1,<x^2>={0},<y^2>={1},<x^2+y^2>={2}".format(x62[N-1][1],y62[N-
1][1],x6jiay62[N-1][1]))
228.     print("β=5,<x^2>={0},<y^2>={1},<x^2+y^2>={2}".format(x63[N-1][1],y63[N-
1][1],x6jiay63[N-1][1]))
229.
230.     plt.show()

```

代码说明：

此处设置 $d=2$ ，并且同时计算 β 为 0.2，1，5 的情况并同步绘制散点图和 $\langle x^2 \rangle$ $\langle y^2 \rangle$ $\langle x^2+y^2 \rangle$ 趋势图。

结果

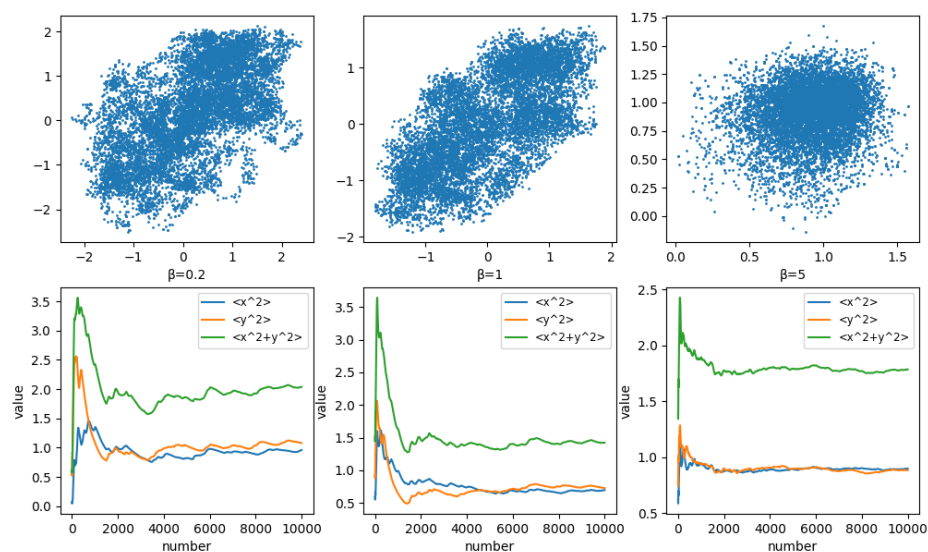


图 1 $d=0.25$

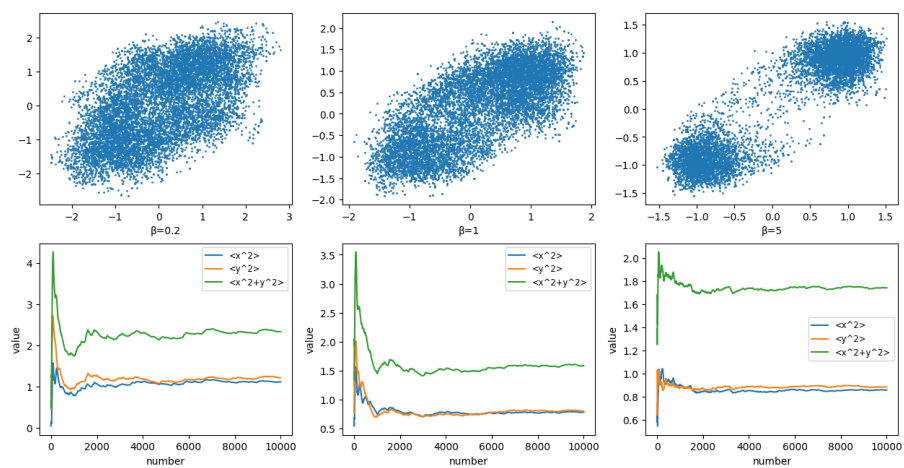


图 2 $d=0.5$

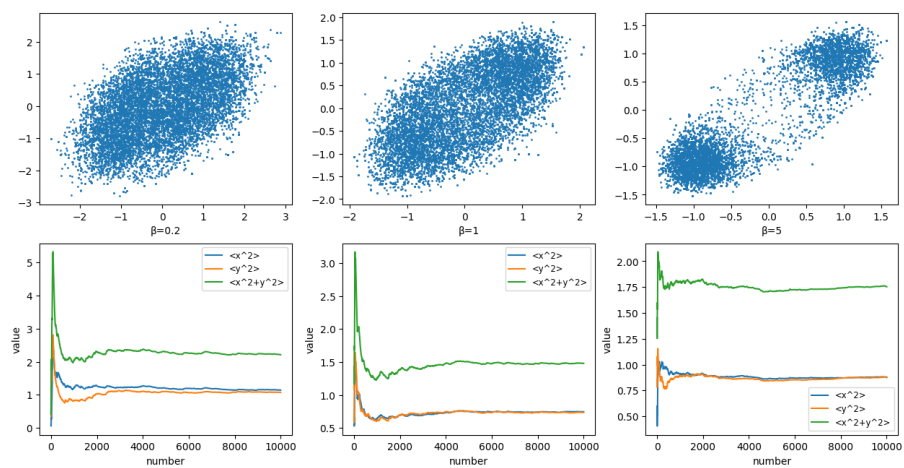


图 3 $d=1$

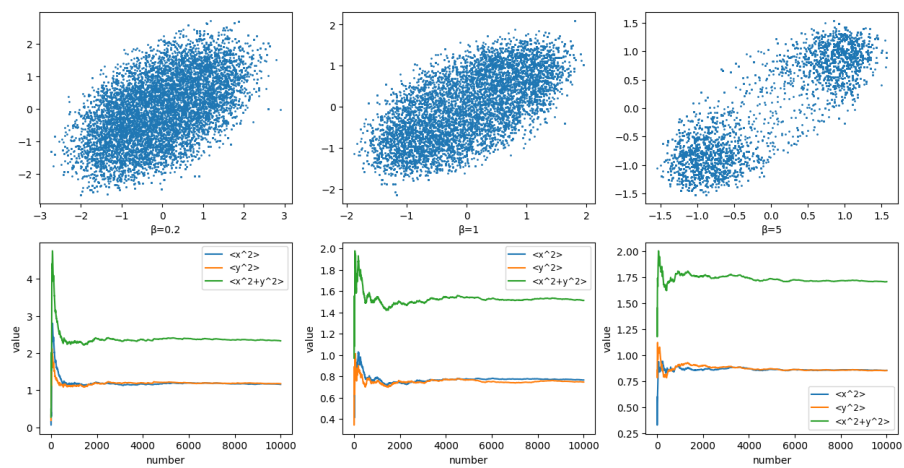


图 4 d=2

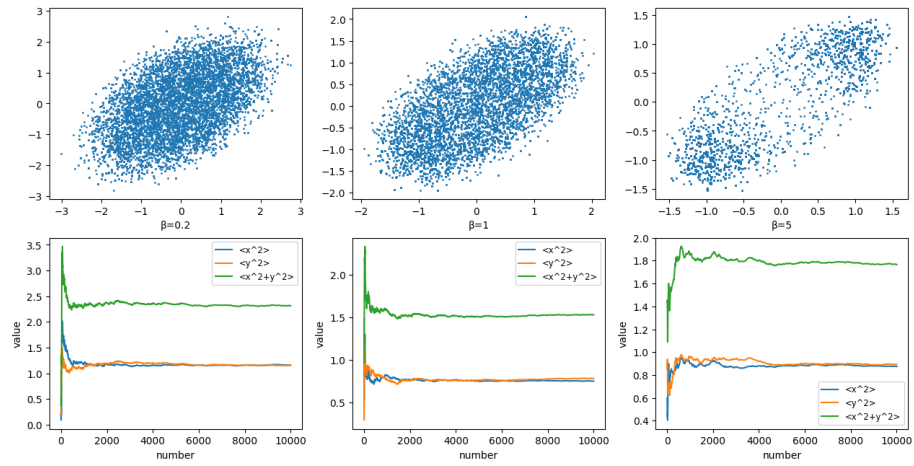


图 5 d=3

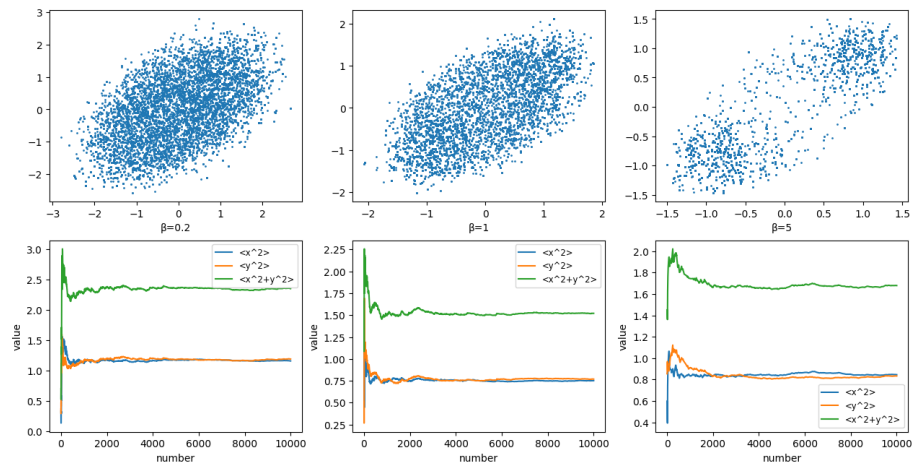


图 6 d=4

打印结果:

d=0.25

$\beta=0.2, \langle x^2 \rangle = 0.960607263047604, \langle y^2 \rangle = 1.078407126175793, \langle x^2 + y^2 \rangle = 2.039014389223397$

$\beta=1, \langle x^2 \rangle = 0.6953775241183783, \langle y^2 \rangle = 0.7264795918013726, \langle x^2 + y^2 \rangle = 1.4218571159197508$

$\beta=5, \langle x^2 \rangle = 0.9000278961667489, \langle y^2 \rangle = 0.8854205297160795, \langle x^2 + y^2 \rangle = 1.7854484258828285$

d=0.5

$\beta=0.2, \langle x^2 \rangle = 1.2226547021959444, \langle y^2 \rangle = 1.1500190780425485, \langle x^2 + y^2 \rangle = 2.372673780238493$

$\beta=1, \langle x^2 \rangle = 0.8294886554314057, \langle y^2 \rangle = 0.7494638301285607, \langle x^2 + y^2 \rangle = 1.5789524855599664$

$\beta=5, \langle x^2 \rangle = 0.8382824743238949, \langle y^2 \rangle = 0.8388798350802324, \langle x^2 + y^2 \rangle = 1.6771623094041272$

d=1

$\beta=0.2, \langle x^2 \rangle = 1.0825660220684636, \langle y^2 \rangle = 1.0630364418131228, \langle x^2 + y^2 \rangle = 2.1456024638815867$

$\beta=1, \langle x^2 \rangle = 0.7228135611059683, \langle y^2 \rangle = 0.7262817698808239, \langle x^2 + y^2 \rangle = 1.4490953309867922$
 $\beta=5, \langle x^2 \rangle = 0.8698341348325056, \langle y^2 \rangle = 0.8561996330427628, \langle x^2 + y^2 \rangle = 1.7260337678752684$

$d=2$

$\beta=0.2, \langle x^2 \rangle = 1.1237816012153046, \langle y^2 \rangle = 1.13811380716408, \langle x^2 + y^2 \rangle = 2.2618954083793845$
 $\beta=1, \langle x^2 \rangle = 0.7508528898281656, \langle y^2 \rangle = 0.7518458197648569, \langle x^2 + y^2 \rangle = 1.5026987095930224$
 $\beta=5, \langle x^2 \rangle = 0.8858883066718742, \langle y^2 \rangle = 0.8809035440181475, \langle x^2 + y^2 \rangle = 1.7667918506900218$

$d=3$

$\beta=0.2, \langle x^2 \rangle = 1.1591079785312624, \langle y^2 \rangle = 1.1547940735386664, \langle x^2 + y^2 \rangle = 2.3139020520699285$
 $\beta=1, \langle x^2 \rangle = 0.7486528985655115, \langle y^2 \rangle = 0.7804318282643625, \langle x^2 + y^2 \rangle = 1.5290847268298742$
 $\beta=5, \langle x^2 \rangle = 0.8738189126730356, \langle y^2 \rangle = 0.8917728876015163, \langle x^2 + y^2 \rangle = 1.7655918002745519$

$d=4$

$\beta=0.2, \langle x^2 \rangle = 1.1576515855683362, \langle y^2 \rangle = 1.1529413833267188, \langle x^2 + y^2 \rangle = 2.310592968895055$
 $\beta=1, \langle x^2 \rangle = 0.7595016539519168, \langle y^2 \rangle = 0.7487534387989684, \langle x^2 + y^2 \rangle = 1.5082550927508853$
 $\beta=5, \langle x^2 \rangle = 0.8705878498615518, \langle y^2 \rangle = 0.8851560334118652, \langle x^2 + y^2 \rangle = 1.755743883273417$

由结果可以得到以下结论：

- 1、随着 β 逐渐变大，表示的温度逐渐减小，散点的分布由**分散于各处**逐渐转为**密集于低势能处**，其对应了退火温度逐渐下降时，允许的势能构型逐渐收敛于最低值。
- 2、随着点数 number 逐渐变大， $\langle x^2 \rangle$ 、 $\langle y^2 \rangle$ 、 $\langle x^2 + y^2 \rangle$ 的值先是有较大波动，后逐渐平稳，最后有小范围波动，这体现出 MCMC 抽样在点数足够大时，最终依分布收敛，以及统计力学的涨落现象。
- 3、对于 d 的变化，可以看到， d 太小 (0.25) 时，达到收敛所需要的抽样次数十分巨大， $\beta=5$ 时更是因为抽样次数过少导致了过拟合。因此 d 的设置应使得抽样时有一定的概率越过势能峰处而到达另一个势能谷，但 d 不能太大，否则会导致新构型被拒绝次数过大，也可能会使收敛变困难，可以设置为所有势能谷之间的最大距离。